

SUJIT SANGRAM SAHOO

Blockchain Security | Assistant Professor

Dept. of CS&IT | Siksha O Anusandhan University | Bhubaneswar – Odisha -751030

IIIT Allahabad | NIT Rourkela | BPUT – Odisha | MNNIT Allahabad

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SUMMARY

I am an Assistant Professor in Department of CS&IT Siksha O Anusandhan University Bhubaneswar, Odisha. I have submitted Ph.D. thesis in the Department Information Technology, Indian Institute of Information Technology Allahabad, India. I received M. Tech. Computer Science from National Institute of Technology Rourkela, India. I received B. Tech. Information Technology from the Biju Patnaik University of Technology Odisha. I am working in the research area of Blockchain, Layer-2 Blockchain, Cryptography, Advance Cryptography, and Network Security. I have 3.8 years of teaching experience and from which 1.5 years of experience from **Motilal Nehru National Institute of Technology Allahabad**.

RESEARCH INTREST

Blockchain Technology

- Scalability Issue in Blockchain.
- Layer-2 Blockchain to make the Consensus free and time management.
- Virtual Payment Model.
- Channel Rebalancing in Bitcoin and Ethereum Network.
- Node Dispute Handling in Virtual State Channels.
- Channel Splitting for Node Idleness in Ledger State Channels.

Cryptography and Security

- Digital Signature Scheme and their modification for Decentralized Networks.
- Certificateless Cryptography for Small Power Device Communication.
- Flood and Loot Attack in Bitcoin Lightning Network.
- Bilinear Pairing Free Cryptography.

EDUCATION QUALIFICATION

Indian Institute of Information Technology Allahabad, India

Ph.D. in Information Technology (**Regular**) (**Thesis Submitted**)

Thesis Title: Instant Payment Systems for Layer-2 Blockchain

CGPA: 9.2

Supervisor: Dr. Vijay Kumar Chaurasiya

August 2018 to Present

Prayagraj, UP, India

National Institute of Technology Rourkela, India

M. Tech. in Computer Science

Specialization: Information security

Thesis Title: Certificateless Nominative Signature based upon DLP

CGPA: 8.05

2014 to 2016

Rourkela, Odisha, India

Biju Patnaik University of Technology Rourkela, India

B. Tech. in Information Technology

CGPA: 7.05

2008 to 2012

Rourkela, Odisha, India

Council of Higher Secondary Education Odisha, India

12th in science

Year: *2006 to 2008*

Board of Secondary Education Odisha, India

10th in Mathematics, Science and English

Year: 2006

TEACHING EXPERIENCE

Siksha O Anusandhan University Bhubaneswar, Odisha, India

1.3 Year

Position: Assistant Professor

Department: Computer Science and Information Technology

Duration: Mar 2023 to Present

Motilal Nehru National Institute of Technology Allahabad, India

1.5 Years

Position: Adjunct Faculty

Department: Computer Science and Engineering

Duration: Aug 2016 to Jan 2018

DharaniDhara Autonomous College Keonjhar Odisha, India

1 Year

Position: Junior Lecturer

Department: Self-financing Department

Duration: July 2013 to Aug 2014

Total: 3.8 years

RESEARCH PUBLICATIONS

IEEE/ACM TRANSACTION: Under Review (2)

SCI / SCIE Journal (Unpaid) Papers: Published (5), Under Review (1)

BOOK CHAPTER: 1 (Published)

CONFERENCE PAPERS: Published (5)

THESIS WORK

Ph. D. Thesis

Indian Institute of Information Technology
Allahabad, India
2018 to Present

Instant Payment Systems for Layer-2 Blockchain

- Summary of Thesis:** One of the fundamental challenges that hinder the further adaption of decentralized cryptocurrencies is scalability. Because current cryptocurrencies require that all transactions are processed and stored on a distributed ledger – the so-called Blockchain – transaction throughput is inherently limited. An important proposal to significantly improve scalability is off-chain protocols, where many transactions are executed without requiring costly interaction with the Blockchain. Examples of off-chain protocols include payment channels and networks currently deployed by popular cryptocurrencies such as Bitcoin and Ethereum. A further extension of payment networks envisioned for cryptocurrencies are so-called state channel networks. Payment channels are an efficient method for performing cheap micropayments in cryptocurrencies. In contrast to traditional on-chain transactions, payment channels have the advantage that they allow for a nearly unlimited number of transactions between parties without involving the Blockchain. But payment networks that only support off-chain payments between users and state channel networks allow the execution of arbitrary complex smart contracts. We introduce an off-chain channel system that offers a new method for connecting channels that is more efficient than the existing “routing transactions” technique over multiple channels. To this end, the method avoids the involvement of the intermediary for each payment. We also consider channel disputes and payment splitting. Moreover, we provide formal security definitions and prove the security of our construction against powerful adversaries. An additional benefit of our structure is channel virtualization, which further reduces latency and costs in complex channel networks.

M. Tech. Thesis

Certificateless Nominative Signature Scheme based upon DLP

National Institute of Technology
Rourkela, India
2014 to 2016

- Summary of Thesis:** The certificateless nominative signature is a concept devoid of the key escrow problem and certificate renewal. In a conventional digital signature scheme, if the digital signature goes unverified by the public

key, the receiver marks it invalid without knowing the message was corrupted, or the private key was false. The sender owns both the public key and the private key. The receiver must pay additional costs to obtain the digital signature certificate. The nominee has a signature with him in the case of a nominative signature. In contrast with the nominative signature, we focus on certificateless public key cryptography, avoiding the use of additional escrow, unlike the identity-based system. Moreover, it does not require any certificate to ensure the keys' authenticity. Our scheme presents a certificateless nominative signature scheme based upon computational hard problems, namely, Discrete Logarithmic Problems and Computational Diffie-Hellman problems. In the proposed method, the partial key is used by a user with the random secret of the user to generate the private and public keys. The nominative signature is publicly verifiable and convertible, and only the nominee can verify the validity of the signature. Moreover, the scheme can withstand key replacement attacks. Also, the proposed scheme is proven to be secure against unforgeability.

TEACHING SKILLS

- **Course Subjects:** Operating System, Cryptography, Data Structures and Blockchain.
- **Programming Languages:** C, C++, Solidity.
- **Operating Systems:** Windows, Linux.

INVITED TALKS AND PRESENTATIONS

Expert Talk: Blockchain on Information Security

Smt. S. R. Patel Engineering College, Gujrat

Sep 2022

IEEE INDICON 2022

IEEE 19th India Council International Conference

Paper Title: Ethereum Compatible Faster Atomic Payment Splitting Network

24th to 26th November 2022

IEEE INDISCON 2022

IEEE 3rd India Council International Flagship Conference

Paper Title: Secure Blockchain Model for Vehicles Toll Collection by GPS tracking: A case study of India

15th to 17th July 2022

CERTIFICATION / INDUSTRIAL WORKSHOP

- IEEE Young Professional Member Since 2020.
- Organizing Four Days Self-Financed Short-term course on “IoT-Machine Learning-Blockchain” Indian Institute of Information Technology Allahabad, India.
- Certified from AICTE Training and Learning (ATAL) Academy FDP on “BLOCKCHAIN” organized by Indian Institute of Information Technology Allahabad, India.
- Certified from AICTE Training and Learning (ATAL) Academy FDP on “Internet of Things” organized by LBS College of Engineering, Kasaragod
- Organizing Five Days Self-Financed Short-term course on “Wireless Network and IoT” Indian Institute of Information Technology Allahabad, India.
- Organizing Eight Days Self-Financed Short-term course on “Network Programming in C/C++” Indian Institute of Information Technology Allahabad, India.
- Certified from AICTE Training and Learning (ATAL) Academy FDP on “BLOCKCHAIN” organized by Excel Engineering College.
- Certified from National Workshop on “Recent Trends in Software Testing”, National Institute of Technology Rourkela, India.
- Certified from National Workshop on “Software Quality Assurance”, National Institute of Technology Rourkela, India.

SCHOLASTIC ACHIEVEMENTS

- MHRD Fellowship for Ph.D., IIIT Allahabad.
- MHRD Fellowship for M. Tech., NIT Rourkela.
- Gate Qualified.
- Best Boarder Award for “Dhirubhai Ambani Hall” NIT Rourkela.
- Working as “General Secretary” in “Dhirubhai Ambani Hall” NIT Rourkela 2015 to 2016.
- I got First prize in Slogan Competition in Vigilance Awareness Program IIIT Allahabad.
- I got state merit Scholarship 2008 to 2012.

RESEARCH PUBLICATIONS

IEEE TRANSACTIONS

- **Sahoo, Sujit Sangram**, and Vijay K. Chaurasiya. "Secure Virtual Non-Atomic Payment Splitting Network on Blockchain." In *IEEE Transaction*. (Under Review)
- **Sahoo, Sujit Sangram**, and Vijay K. Chaurasiya. "Prevention of Flood and Loot attack with secure computing." In *ACM Transactions on Internet Technology*. (Under Review)

SCI / SCIE JOURNALS

- **Sahoo, Sujit Sangram**, Mahesh Mohan Hosmane, and Vijay Kumar Chaurasiya. "A secure payment channel rebalancing model for layer-2 blockchain." *Internet of Things* 22 (2023): 100822.
- **Sahoo, Sujit Sangram**, and Vijay Kumar Chaurasiya. "EASB: ECC based aggregate signature without bilinear pairing for blockchain." *Multimedia Tools and Applications* (2023): 1-20.
- **Sahoo, Sujit Sangram**, Aravind R. Menon, and Vijay Kumar Chaurasiya. "Blockchain Based n-party Virtual Payment Model with Concurrent Execution." *Arabian Journal for Science and Engineering* (2023): 1-28.
- **Sahoo, Sujit Sangram**, and Vijay Kumar Chaurasiya. "VIBE: blockchain-based virtual payment in IoT ecosystem: a secure decentralized marketplace." *Multimedia Tools and Applications* (2023): 1-26.
- **Sahoo, Sujit Sangram**, and Vijay Kumar Chaurasiya. "Proof of location-based delivery system using multi-party virtual state channel: a blockchain model." *The Journal of Supercomputing* (2023): 1-31.
- **Sahoo, Sujit Sangram**, and Vijay K. Chaurasiya. " Deterministic Nonce Generation to Prevent Key Exfiltration Attack." In *Wireless Personal Communication*. (Under Review)

BOOK CHAPTERS

- Kumar, Akash, **Sujit Sangram Sahoo**, and Vijay Kumar Chaurasiya et al. "Secure Certificateless Maximum Achievable Throughput in Successive IoT Relay Networks." In *Information Technology Security: Modern Trends and Challenges*, pp. 189-208. Singapore: Springer Nature Singapore, 2024.
- **Sahoo, Sujit Sangram**, Aravind R. Menon, and Vijay K. Chaurasiya. " Secure and Authenticate Decentralize Model for Vehicle on Blockchain Off-Chain Payment System." In *Blockchain in Smart Networks*. (Communicated)

CONFERENCES

- **Sahoo, Sujit Sangram**, Aravind R. Menon, and Vijay K. Chaurasiya. "Secure Blockchain Model for Vehicles Toll Collection by GPS tracking: A case study of India." In *2022 IEEE India Council International Subsections Conference (INDISCON)*, pp. 1-6. IEEE, 2022.
- **Sahoo, Sujit Sangram**, Mahesh M Hosmane, Aravind R. Menon, and Vijay K. Chaurasiya. "Ethereum Compatible Faster Atomic Payment Splitting Network." In *2022 IEEE India Council International Subsections Conference (INDICON)*. IEEE, 2022.
- **Sahoo, Sujit Sangram**, Mahesh M Hosmane and Vijay K. Chaurasiya. "Blockchain based Marketplace for Farmers using PERUN Payment System." In *2022 International Conference on Emerging Global Trends in Engineering and Technology*, 2022.
- Mohanty, Sujata, **Sujit Sangram Sahoo**, and Banshidhar Majhi. "Certificateless nominative signature scheme based upon DLP." In *2016 International Conference on Electrical, Electronics, and Optimization Techniques (ICEEOT)*, pp. 1241-1245. IEEE, 2016.
- Aditia, Mayank and **Sujit S. Sahoo** et.al. "Optimized CL-PKE with lightweight encryption for resource constrained devices." In *Proceedings of the 20th International Conference on Distributed Computing and Networking*, pp. 427-432. 2019.

DATE:

PLACE:

SUJIT SANGRAM SAHOO